README – Analysis Code Submission

Thesis: Melanoma incidence in Norway, 1980–2024: Diverging age and sex trends

This folder contains the R scripts submitted as documentation of the analytical work performed for the accompanying master's thesis, *Melanoma incidence in Norway, 1980–2024: Diverging age and sex trends*. The submitted scripts represent the final documented analysis workflow used to generate the results presented in this thesis. During the course of the project, the analysis code evolved through multiple iterations that included exploratory analyses, intermediate data checks, alternative plotting approaches, and local testing. For submission, the scripts were reviewed and cleaned by removing redundant code, exploratory commands, duplicated sections, and local computer paths, while preserving the analytical procedures, coding style, and workflow used for the final analyses.

Scope of the submitted code

The submitted scripts cover the principal analytical components of the thesis, including:

- descriptive analyses and preparation of Table 1;
- birth cohort and calendar period analyses;
- regional analyses;
- anatomic site and histological subtype summaries;
- contextual immigration analyses;
- preparation of manuscript figures and supplementary material.

The Annual Percent Change (APC) and Average Annual Percent Change (AAPC) estimates reported in the manuscript were calculated using the National Cancer Institute Joinpoint Regression Program. The exported Joinpoint results were subsequently imported into R for data preparation, organisation, comparison across strata, preparation of tables, and generation of figures.

Notes on final figure layout

Final panel labels, legend placement, spacing, and overall figure layout were adjusted manually after export to ensure consistency with the target journal's formatting requirements. The annual fitted incidence-rate curves presented in the supplementary appendix were produced directly in the Joinpoint Regression Program and were therefore not recreated in R.

Software and reproducibility

Data preparation, descriptive analyses, data management, and figure generation were performed in R version 4.5.1 (R Foundation for Statistical Computing, Vienna, Austria). Joinpoint regression analyses (APC and AAPC) were conducted using the Joinpoint Regression Program version 5.4.0 (National Cancer Institute, Bethesda, MD, USA).

The principal R packages used throughout the analyses included: dplyr , tidyr , readxl, ggplot2, openxlsx, stringr.

The submitted scripts use relative file paths and are organised into separate analysis modules to facilitate revision and reuse. As no analyses involving random sampling, resampling, bootstrapping, or simulation were performed, setting a random seed was not required for the analytical workflow.

Folder structure and script execution

All file paths are relative. The original registry data obtained from the Cancer Registry of Norway are not included because they are subject to data-use restrictions. The project structure is organised as follows:

```
R_Code/
│
├── README.pdf
├── 00_setup.R
├── 01_Descriptives.R
├── 02_Cohort_Analysis.R
├── 03_Period_Analysis.R
├── 04_Regional_Analysis.R
├── 05_Plots.R
└── 06_Immigration.R
```

Scripts should be run in numerical order, as later scripts depend on objects and outputs generated by previous scripts.

Variable definitions

The submitted scripts use standardised variable names to improve readability. Where necessary, original variable names from the source datasets were harmonised (e.g. Gender → sex). The principal analysis variables are:

Variable	Description
year	Calendar year of diagnosis
sex	Male or female
age_group	10-year age groups (10–29, 30–39, 40–49, 50–59, 60–69, 70–79, and 80+ years)
region	Norwegian health region (South-Eastern, Western, Central, Northern)
cases	Number of incident melanoma cases
population	Population denominator used for incidence calculations
incidence_rate	Age-standardised incidence rate per 100,000 person-years (European Standard Population 2013 where applicable)
cohort	Approximate birth cohort derived by subtracting the age-group midpoint from calendar year

Notes on the analyses

- Birth cohorts were approximated by subtracting the midpoint of each age group from calendar year.
- APC and AAPC estimates were calculated using the Joinpoint Regression Program and subsequently processed in R.
- Regional analyses were performed using the four Norwegian health regions.
- Anatomic site and histological subtype analyses were restricted to the coding periods used in the thesis.
- Supplementary Table S3 was prepared using published percentages from Statistics Norway for 1980, 1990, 2000, 2010, 2020, and 2024.